

Whole Mount Subcutaneous White Adipose Tissue Immunostaining

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Reagents:

Fixation

2% PFA in 1X PBS

Wash Buffers

1X PBS

1X PBS w/ 10U/mL Heparin

1X PBS w/ 0.02% NaN₃ (optional; to prevent bacterial growth)

Block

Blocking Solution

1X PBS

2.5% BSA

1% Triton X-100

Autofluorescence Quenching Agent

Typogen Black

0.1% Typogen Black in 70% EtOH

TrueBlack

1X TrueBlack made from 20X Stock. Dilute with 70% EtOH

Hydrogen Peroxide Bleaching

5% H₂O₂ in 20% DMSO/Methanol

Antibody Solutions

Primary Antibody Solution

Concentration is antibody dependent. Typically, 1:250 – 1:1000

Diluted in 1X PBS

Secondary Antibody Solution

Concentration is antibody/fluorophore dependent. Typically, 1:500 – 1:1000

Diluted in 1X PBS

Mountant

Glycerol Mounting Medium w/ DABCO

Tools and Equipment:

- 6 well plate
- Flat forceps
- Plate rotator in cold room 4°C
- Large glass slides (75x51mm)
- Large glass coverslips (48x60mm)
- Medium to large binder clips
- Micropipettes

Fixation Procedure:

1. Mark the top a 12 or 6 well plate with the IDs of the animals and types of adipose tissue that will be collected so that each tissue sample will have its own well and the ID above it when the plate cover is on the plate.
2. Place 2-5mL of cold 2% PFA in each well and place dissected tissue into the appropriately marked well.
3. Place tissues in appropriate wells and incubate in the fridge for 4 -16 hrs) depending in tissue thickness (i.e. a non-obese mouse with scWAT tissue of no more than ~0.20g needs only 4hrs a fat depot weighing 0.50g needs approx. 8hrs, ~1.0g 16hrs).
4. After the appropriate fixation incubation remove the 2% PFA, rinse twice with 1X PBS/10U/ml Heparin and mildly agitate on rotator 30min each rinse, in cold room.
 - a. PBS w/ Heparin is used to rinse blood from vessels to reduce tissue autofluorescence.

Squishing and Blocking Procedure:

1. On a glass slide write the information for your tissue sample so that it will be identifiable to you after it is squished.
2. Remove 12 or 6 well plate containing tissue samples from cold room.
3. Place a tissue on large glass slide and add a few more drops of 1X PBS onto the tissue.
4. Place a second slide on top of the sample and press slides together (with the adipose tissue in-between them) using a binder clip. Use 2 binder clips on larger tissues to increase the effectiveness of compression.
5. Place in fridge for 1hr 30min. Longer compression times can result in thinner tissues and may be necessary for large >0.50g tissues. Be careful not to let tissues dry out.
 - a. It may be necessary to rotate the binder clip to place force on other side of tissue halfway through squishing time.
6. After squishing is complete perform blocking for non-specific binding. Add 2-5 mL of blocking solution (enough to cover the tissue) to each well.
7. Return 6-12 well plate to rotating shaker in cold room. Incubate the tissue samples in blocking solution overnight. Tissues can be left in blocking solution longer (up to 7 days) if blocking solution is replaced daily.
8. Next, perform one of the three following autofluorescence quenching methods:
 - a. Typogen Black: Used to block autofluorescence in all but far-red channels. Increases far-red autofluorescence. Not suitable for 647nm fluorophore unless bound to something extremely highly expressed (e.g. Isolectin IB₄)
 - i. Place tissues back into the 6 well plate with about 5 mL of 0.1% Typogen Black to reduce tissue autofluorescence [1]. Make sure that there is

enough in each well to submerge the tissue sample. If there is not add more. Place on the rotator at room temp for 20 min, for mild agitation.

- ii. After 20 min remove Typogen Black from each well and add 5mL 1X PBS w/Heparin (10U/ml) (to be used as wash buffer) to each well. Place on rotator at RT for 1hr, rinse and repeat.
 - iii. The next 2 washes should be done on a rotator at 4°C (or until there is no run-off, may need overnight washing.)
 - iv. Move onto IF staining.
- b. True Black: Used to block autofluorescence in all channels including far-red. Results in reduced signal intensity so it should only be used when 647nm fluorophore is necessary. Also, True Black is much more expensive than other quenching methods.
- i. Place tissues back into the 6 well plate with about 5 mL of 1X (diluted from 20X w/ 70% EtOH) True Black to reduce tissue autofluorescence [1]. Make sure that there is enough in each well to submerge the tissue sample. If there is not add more. Place on the rotator at room temp for 20 min, for mild agitation.
 - ii. After 20 min remove Typogen Black from each well and add 5mL 1X PBS w/Heparin (10U/ml) (to be used as wash buffer) to each well. Place on rotator at RT for 1hr, rinse and repeat.
 - iii. The next 2 washes should be done on a rotator at 4°C (or until there is no run-off, may need overnight washing.)
 - iv. Move onto IF staining.
- c. Hydrogen Peroxide Bleaching: Quenching method used in many clearing protocols. Results in reduced autofluorescence across all channels without reducing signal intensity. Unfortunately, this method adds a day and several additional hours to the protocol. Adapted from Renier et al. 2014 [2].
- i. Incubate tissues in 50% MeOH in 1X PBS for 1hr at RT
 - ii. Incubate tissues in 80% MeOH in 1X PBS for 1hr at RT
 - iii. Incubate tissues in 80% MeOH in 1X PBS for 1hr at RT, 2 times
 - iv. Bleach with 5% H₂O₂ in 20% DMSO/MeOH overnight at 4°C
 - v. Incubate tissues in 100% MeOH in 1X PBS for 1hr at RT, 3 times
 - vi. Incubate tissues in 20% DMSO/MeOH for 1hr at RT, 2 times
 - vii. Incubate tissues in 80% MeOH in 1X PBS for 1hr at RT
 - viii. Incubate tissues in 50% MeOH in 1X PBS for 1hr at RT
 - ix. Incubate tissues in 1X PBS for 1hr at RT
 - x. Move onto IF staining

IF Staining:

NOTE: from now on 1X PBS without heparin should be used as wash buffer. To prevent bacterial growth, 1X PBS w/ 0.02% NaN₃ can be used instead.

Primary antibody

1. Prepare primary antibody solution (5mL per tissue sample plus at least 5 mL excess), vortex to mix.
 - a. Diluent is 1X PBS
 - b. Concentrations of primary antibodies are antibody specific
For example: 10mL of 1:250 dilution of antibody solution would be made from:
40µL anti PGP9.5
9960µL 1x PBS
2. Add 2-5mL of primary antibody solution to each well containing a tissue sample so that each tissue is submerged in primary antibody solution. If there is not enough solution add more so that the tissue samples are submerged.
3. Return plate to cold room on plate rotator and incubate there for 48hrs.

Wash and secondary antibody

1. After the 48hr incubation with primary antibodies remove plate from the cold room and use a 1mL pipette to remove the primary antibody solution from the well containing tissue.
2. Add 5mL 1X PBS (to be used as wash buffer) to each well containing adipose samples on the plate and return to the rotator in the cold room for 1 hour.
3. Repeat steps 1 and 2 for a total of 4-6 hours
4. Prepare secondary antibody solution (5mL per tissue sample plus at least 5 mL excess), vortex to mix.
 - a. Diluent is 1x PBS
 - b. Secondary antibody concentration is antibody and fluorophore specific
5. Remove and discard 1X PBS with 1mL pipette.
6. Add 5mL of secondary antibody solution to each well containing a tissue sample so that each tissue is submerged in secondary antibody solution. If there is not enough solution add more so that the tissue samples are submerged.
7. Wrap the plate in tin foil to protect the samples from light.
8. Return plate to cold room on plate rotator and allow to incubate there overnight.
9. After the overnight incubation with the secondary antibody solution remove plate from the cold room and use a 1mL pipette to remove secondary antibody solution from the wells containing tissue.
10. Add 5mL 1x PBS (to be used as wash buffer) to each well containing tissue samples on the plate and return to the rotator in the cold room for 1 hour, wrapped in foil.

11. Repeat steps 1 and 2 for a total of 4-6 hours
12. Optional: Repeat IF staining procedure for additional antibodies
13. Optional: at this time proceed to the Whole Depot Isolectin Staining Protocol if vasculature stain is desired.
14. Continue to mounting procedure.

Mounting

1. After washing for 4-6 hours set up an area to mount adipose tissues onto slides.
 - a. Work on top of a large kim wipe.
2. Remove the 6 well plate with the tissue samples from the cold room and keep protect the samples from light as much as possible.
3. On a large glass slide right down the identifying data for the tissue sample to be mounted.
4. Remove tissue sample from the 6 well plate using forceps and place on large glass slide.
 - a. Return the 6 well plate to the drawer to protect the other tissue samples from light.
5. Use a dry kimwipe to dab the excess 1x PBS from the slide.
6. Add a 3-4 drops of glycerol mountant to the slide.
7. Cover the tissue sample with a cover slip large enough to cover the entire sample.
8. Push *gently* down so that the mounting fluid is evenly dispersed with few bubbles under the cover slip.
9. Repeat for all tissues.
10. Place flat weights on top the tissues (several large textbooks work well) and leave weight on for 3 days -- squish time depends on tissue size/thickness.
11. After 3 days carefully remove the weights
12. Use nail polish to seal the edges of the cover slip to the slide.
13. Place in light protective box cover side up in a drawer to dry and repeat for all slides.
14. Allow slides to dry covered and in a drawer overnight
15. Proceed to imaging and analysis.

References

1. Schnell, S.A., W.A. Staines, and M.W. Wessendorf, *Reduction of Lipofuscin-like Autofluorescence in Fluorescently Labeled Tissue*. Journal of Histochemistry & Cytochemistry, 1999. **47**(6): p. 719-730.
2. Renier, N., et al., *iDISCO: a simple, rapid method to immunolabel large tissue samples for volume imaging*. Cell, 2014. **159**(4): p. 896-910.